

Yarone Meir Tokayer

Updated: July 15, 2026

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EDUCATION

Yale University

Ph.D., Physics, Advisor: Frank van den Bosch

Thesis: “Probing collisional gravitational systems with the gravothermal fluid model”

Certificate in Public Communication

M.S. and M.Phil

New Haven, CT

exp. 2027

Apr. 2026

May 2023

Columbia University

M.A., Philosophical Foundations of Physics, Advisor: David Z. Albert

Thesis: “Probability in Everettian Quantum Mechanics”

New York, NY

Feb. 2020

The Cooper Union

B.S., Engineering, Minor in Mathematics, *summa cum laude*

Senior Project: “Muscle Denervation and Neurotensin as a Therapeutic Target for ALS”

New York, NY

May 2014

SKILLS AND LANGUAGES

- **Programming languages:** Python, MATLAB, C++, HTML/CSS, FORTRAN
- **Software & tools:** NASA HEASoft, Mathematica, Latex, LabVIEW, Excel
- **N-body code:** *gyrfalcON*, *Agama*
- **Python Libraries:**
Astronomy: *Astropy*, *Stingray*, *PyXSPEC*
Data and Visualization: *Pandas*, *NumPy*, *Matplotlib*
- **Telescope data:** *Chandra*, *Swift*, *NuSTAR*, *NICER*
- **Ground telescope observing experience:**
Palomar: Double Spectrograph, TripleSpec
Keck: NIRC2+AO, OSIRIS+AO
- **Design tools:** Arduino, Microchip PIC, AutoCAD, SolidWorks, laser cutting
- **Spoken languages:** English (native), Hebrew (fluent), Yiddish (basic), German (basic)

RESEARCH EXPERIENCE

Modeling the dynamical structure of dark matter halos

Graduate Research Assistant, PI: Frank van den Bosch

Theoretical and computational astrophysics.

- Developed gravothermal fluid codes to model stellar mass segregation and *SIDM* core collapse.
- Implemented idealized *N*-body code to model *CDM* halo growth and compare with cosmological predictions.

Jun. 2024–Present

Yale University

Quantifying Selection Effects in Deep-Field AGN Surveys Using BASS

Graduate Research Assistant, PIs: Michael Koss, C. Megan Urry

Extra-galactic high energy astrophysics.

- Developed simulation data pipeline to generate and fit high-*z* *Chandra* AGN spectra.
- Spectral analysis of 2800 simulated spectra to quantify *Chandra* bias.
- Work published in *ApJ*; poster presented at the Jan. 2023 AAS meeting.

Jul. 2021–Aug. 2024

Yale University

Constraining Dark Matter Profiles with GSSL Measurements

Graduate Research Assistant, PI: Priyamvada Natarajan

Data-driven cosmology.

Jan. 2022–May 2024

Yale University

- Developed fitting algorithms for cluster DM subhalo profiles from lensing measurements.
- Derived empirical c-M relation, furthering Λ CDM tension.
- Work published in *ApJ*.

Galactic X-ray astronomy data analysis

Mar. 2020–Dec. 2020

Research Assistant, PIs: Charles Hailey, Kaya Mori

Columbia University

Galactic high energy astrophysics.

- Timing analysis of NICER observations of AR Scorpii. Sub- μ s precision spin measurement; found pulsed non-thermal emission in X-ray band.
- Imaging analysis of NuSTAR observations of the “Eel Nebula.” Discovered synchrotron burnoff effect in the PWN and evidence of a shock feature. Work published in *ApJ*.
- Spectral and timing analysis of NuSTAR observations of HESS J0632+057. Work published in *ApJ*.

Assembly and testing of detector modules for GAPS experiment

Aug. 2019–Feb. 2020

Research Assistant, PI: Charles Hailey

Columbia University

Experimental particle astrophysics.

- Fabrication, testing, and assembly of Si(Li) detector array modules for GAPS flight (2022); aims to detect antimatter evidence of DM annihilation in the galactic halo.

ALS pathology in mouse and cell culture models

Sep. 2013–May 2014

Research Assistant, PI: Christopher Henderson; Supervisor: Dima Yudin

Columbia Univ. Med. Ctr

Molecular and Cellular Neurobiology.

- Senior capstone project: ALS pathology in mouse and cell culture models. Immunohistochemistry to measure denervation over time. Optimization of embryonic stem cell-derived motor neuron cultures to test trophic factors.

Lung Perfusion Bioreactor

Jan. 2011–May 2011

Undergraduate Researcher, PIs: Eric Lima, Gordana Vunjak-Novakovic

Cooper Union/Columbia University

- Designed and built prototype used for testing on swine lung.

Med. Ctr

PUBLICATIONS AND POSTERS

A complete list of publications can also be found on [Google Scholar](#).

Refereed Journal Publications

1. Tokayer, Y. M., Koss, M., Urry, C. M., et al., *ApJ*, 982, 134 (2025).
doi:[10.3847/1538-4357/adb8c9](https://doi.org/10.3847/1538-4357/adb8c9)
2. Tokayer, Y. M., Dutra, I., Natarajan, P., et al., *ApJ*, 970, 143 (2024).
doi:[10.3847/1538-4357/ad51fd](https://doi.org/10.3847/1538-4357/ad51fd)
3. Guolo, M., Gezari, S., Yao, Y., et al. (incl. Tokayer, Y. M.), *ApJ*, 966, 160, (2024).
doi:[10.3847/1538-4357/ad2f9f](https://doi.org/10.3847/1538-4357/ad2f9f)
4. Burgess, D., Mori, K., Gelfand, J. D., et al. (incl. Tokayer, Y. M.), *ApJ*, 930, 148, (2022).
doi:[10.3847/1538-4357/ac650a](https://doi.org/10.3847/1538-4357/ac650a)
5. Tokayer, Y. M., An, H., Halpern, J. P., et al., *ApJ*, 923, 17, (2021).
doi:[10.3847/1538-4357/ac2c6a](https://doi.org/10.3847/1538-4357/ac2c6a)

Conference Posters

6. Tokayer, Y. M., Koss, M., Urry, C. M., et al., Quantifying Selection Effects in Deep-Field AGN Surveys with BASS. In: *241st Meeting of the AAS*, January 8–12, 2023. Seattle, WA.
[Link to poster](#).

7. Woo, J., An, H., Burgess, D., et al. (incl. **Tokayer, Y. M.**), Multi-wavelength Study of PeVatron Candidate Pulsar Wind Nebulae. In: *AAS/High Energy Astrophysics Division 19th Annual Meeting*, March 13–17, 2022. Pittsburgh, PA.
8. Saffold, N., **Tokayer, Y. M.**, Mori, K., A NICER X-ray View of White Dwarf Pulsar AR Scorpii. In: *237th Meeting of the AAS*, January 10–15, 2021. Virtual.
[Link to poster](#).

ACADEMIC PRESENTATIONS

Invited Talks

9. “Core-collapsed SIDM halos as massive SMBH seeds,” Hebrew University Cosmolunch; Apr. 24, 2025. Jerusalem, Israel.
10. “The Unified AGN Model in X-ray observations,” Columbia University High-Energy Astrophysics Meeting; Jan. 28, 2022. Virtual.

Conference Talks

11. “Gravothermal Evolution with Multiple Mass Species,” [Modelling and Observing Dense Star Clusters in Vietnam Dense Stellar Systems from Globular to Nuclear Star Clusters \(MODEST26\)](#); Aug 16-22, 2026; Quy Nhon, Vietnam.
12. “Core-collapsed SIDM halos as massive SMBH seeds,” [YCIU Workshop](#); May 30, 2025; New Haven, CT.
13. “Using BASS to Detect Obscuration Bias and Test AGN Fitting Models for Low Count Data,” BASS2024: New Horizons for Understanding Nearby AGN; April 22, 2024; Virtual.
14. “Quantifying AGN Selection Effects in the Chandra COSMOS-Legacy Survey with BASS,” [Accretion History of AGN \(AHA\) III Workshop](#); December 15, 2023; Miami, FL.
15. “Quantifying Selection Effects in Deep-Field AGN Surveys,” [New England Regional Quasar and AGN Meeting \(NERQUAM\)](#); May 6, 2022; Storrs, CT.
16. Saffold, N., **Tokayer, Y. M.**, “A NICER X-ray View of White Dwarf Pulsar AR Scorpii,” [Spring 2021 NICER Data Analysis and Science Workshop](#); May 13, 2021. Virtual.
[Video of presentation](#).

ACCEPTED PROPOSALS

17. Co-Investigator. Ref #GN-2024A-Q-138 for Gemini GMOS Observations. *Resolving Mrk 248: A Potential Triple AGN with a Hidden 500 pc Dual AGN*.
PI: Michael Koss
18. Co-Investigator. Proposal 8087 for NuSTAR Observations. *A Survey of the Most Luminous Hard X-ray Selected Obscured Quasars at $z=0.2-0.4$* .
PI: Michael Koss
19. Co-Investigator. Proposals 90296 and 094349 for XMM-Newton Observations. *A Survey of the Most Luminous Hard X-ray Selected Obscured Quasars at $z=0.2-0.4$* .
PI: Michael Koss

CONFERENCES, WORKSHOPS, AND SUMMER SCHOOLS ATTENDED

- **Modelling and Observing Dense Star Clusters in Vietnam Dense Stellar Systems from Globular to Nuclear Star Clusters (MODEST26)**
August 16–22, 2026. Quy Nhon, Vietnam.
- **AI Winter School**
Brown University. January 6–9, 2026. Virtual.
- **Yale Center for the Invisible Universe (YCIU) Workshop**
Yale University. May 30, 2025. New Haven, CT.
- **Accretion History of AGN (AHA) III Workshop**
University of Miami. December 14–17, 2023. Miami, FL.
- **Cosmology Summer School**
University of Michigan. June 5–9, 2023. Ann Arbor, MI.
- **241st Meeting of the AAS**
January 8–12, 2023. Seattle, WA.
- **New England Regional Quasar and AGN Meeting (NERQUAM)**
University of Connecticut. May 6, 2022. Storrs, CT.
- **NICER Data Analysis and Science Workshop**
May 13, 2021. Virtual.

TEACHING

Leitner Observatory and Planetarium

Planetarium Presenter

January 2024–Present

- Write and present planetarium shows using night sky simulation for groups spanning pre-K to adult
- Design and deliver classroom presentations about astronomy and outer space for school groups K-12
- Run telescope demonstrations for evening groups

Yale University

Course design

- Designed, wrote lecture slides and problem sets for undergraduate course in *Cosmology (ASTR 170)*; Instructor: Priyamvada Natarajan Summer/Fall 2024

Graduate Teaching Fellow

- Graduate Cosmology (PHYS 6000) Spring 2026
- Introduction to Mathematical Methods of Physics (PHYS 301, PHYS 4000) Fall 2022, Fall 2025
- Graduate Statistical Physics I (PHYS 512) Spring 2024, Spring 2025
- Graduate Classical Mechanics (PHYS 500) Fall 2024
- University Physics for the Life Sciences (PHYS 171) Spring 2023
- General Physics Laboratory II (PHYS 166L/S166) Spring 2022, Summer 2022
- General Physics Laboratory I (PHYS 165L) Fall 2021

SAR High School

Chemistry Teacher

Spring 2021

- Taught four 10th grade chemistry sections spanning three levels from remedial to honors

Physics Teacher and Advisor

Fall 2014–Spring 2019

- Designed curriculum and taught 4 physics courses on two tracks for 11th and 12th grade
- Taught 11 grade Jewish Philosophy course, 9 and 11 grade Jewish text study
- 10th grade advisor
- Designed and oversaw engineering-related elective and co-curricular programming
- Coached robotics team that won second place in [robotics competition](#) at the Technion in Haifa, Israel

Naaleh High School for Girls

STEM Teacher

Fall 2019–Spring 2020

- Wrote curriculum and taught a course in computer programming and engineering design
- Designed and instructed Python coding co-curricular; club was a member of [Girls Who Code](#)

The Cooper Union

Teaching Assistant

Fall 2011

- Introductory Physics Lab (PH291)

COMMUNITY ENGAGEMENT AND OUTREACH

Service

Graduate Student Mentor , Yale Undergraduate Mentorship Initiative	2025–Present
Co-Founder and Organizer , History and Foundations of Physics Reading Group	2022–Present
Board of Trustees , Slifka Center for Jewish Life at Yale	2023–2024
Graduate Student Representative , Physics Department Faculty Hiring Committee	Spring 2023

Outreach

Astronomy on Tap	Fall 2022
<i>-Astronomy nights at a New Haven bar, featuring trivia and talks by members of the Yale Astronomy Department.</i>	
Super Science Showdown , Yale Open Labs	Spring 2022
<i>-Interactive science events for students in grades 6-8 in New Haven County.</i>	
Engineers as Teachers , Iridescent and Cooper Union	Spring 2011
<i>-Wrote and built interactive lesson plans on the topic of sound and music</i>	
<i>-Lessons were taught at family science nights in local NYC middle school</i>	

Public Talks

1. “Are we alone? The new search for other worlds and habitable planets beyond the Solar System,” SAR High School; Apr. 17, 2024; Riverdale, NY.
[Video of talk.](#)
2. “The hype is real: what we are already learning from the most powerful telescope in human history,” SAR High School; Jan. 4, 2023; Riverdale, NY.
[Video of talk.](#)

AWARDS AND SCHOLARSHIPS

- **3-Minute Thesis Competition Finalist**
Yale University Graduate School of Arts and Sciences, 2026
- **Leigh Page Award for Excellence in Graduate Student Teaching**
Yale University Physics Department, 2025
- **Teacher Award**
2017 Robotraffic Competition at the Technion in Haifa, Israel, as coach to the SAR High School team
- **Entrance Scholarship**
Philosophical Foundations of Physics Program, Columbia University, Fall 2016

- **Tau Beta Pi**
Engineering Honors Society
- **Goodman Prize**
Cooper Union, Spring 2013
- **Dean's List**
Cooper Union, all semesters